

PNS ERP — Learning & Reference Guide

Project: TallyPNS / PNS ERP
Repo: <https://github.com/vrajput032/TallyPNS>
Local path: ~/Documents/Telly
Guide date: 24 Jul 2026

Keep this PDF private. It lists accounts, URLs, and env layout.
Secrets (passwords, JWT keys, API keys, deploy hooks) live only in local .env files — never commit them to Git.

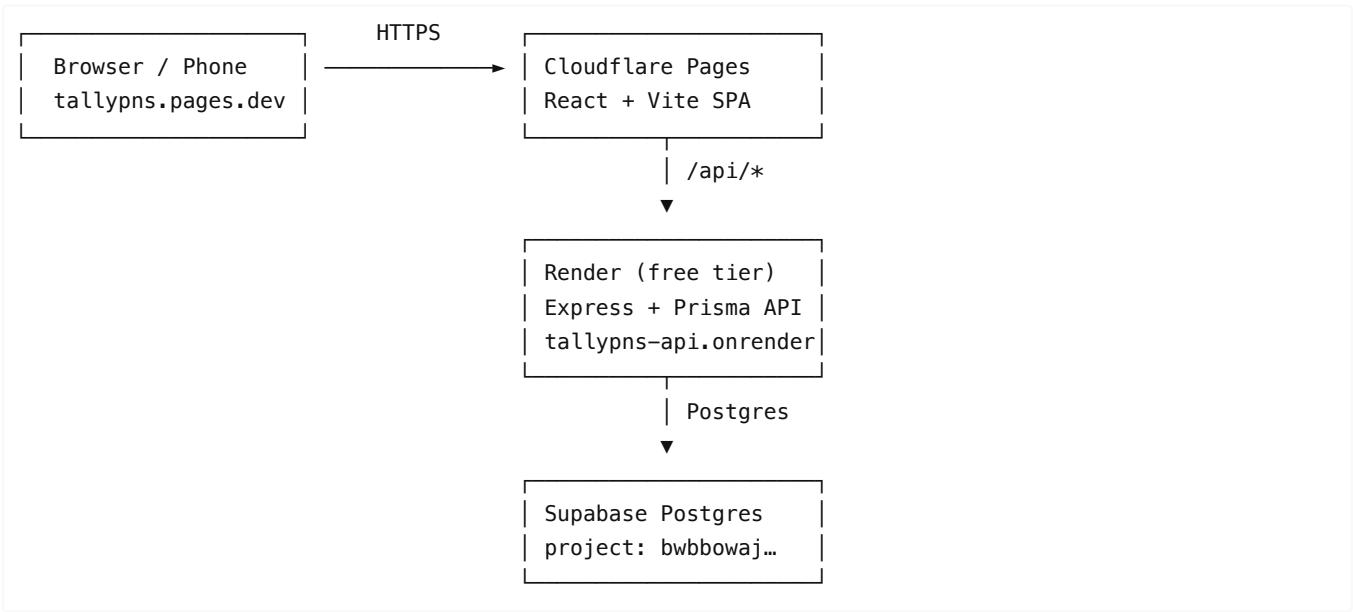
1. What is this project?

PNS ERP is a small business ERP for **PNS Enterprises** (Sonipat, Haryana):

- Customers, vendors, products
- Sales invoices & purchase bills (GST tax invoice print)
- Inventory / stock
- GST summary & reports
- Cash / Bank modules (placeholders for later)

Think of it as a lightweight Tally-like web app for one company.

2. Big picture architecture



Layer	Technology	Where it runs
Frontend	React 19, Vite, TypeScript, Tailwind 4, shadcn/ui	Cloudflare Pages
Backend	Express, TypeScript, Prisma, Zod, JWT	Render
Database	PostgreSQL	Supabase (Tokyo / ap-northeast-1)
Auth	Email + password, JWT access + refresh	Backend
Hosting DNS	GitHub repo → Render + Cloudflare	—

3. Production & local URLs (memorize)

What	URL
Live app (frontend)	https://tallypns.pages.dev
Live API (backend)	https://tallypns-api.onrender.com
API health check	https://tallypns-api.onrender.com/health
API base (used by app)	https://tallypns-api.onrender.com/api
Local frontend	http://localhost:5173
Local backend	http://localhost:4000
Local API base	http://localhost:4000/api
Supabase dashboard	https://supabase.com/dashboard/project/bwbbowajgazuljbxdwng
Supabase project URL	https://bwbbowajgazuljbxdwng.supabase.co
GitHub	https://github.com/vraiput032/TallyPNS
Render service	tallypns-api (Ohio, free)
Cloudflare Pages project	tallypns

4. Accounts & logins

App admin (seeded)

Field	Value
Email	admin@pnsenterprises.com
Password	admin123
Name	Admin
Role	ADMIN

Seed script: `backend/prisma/seed.ts`

Command: `npm run seed -w backend`

Company printed on invoices

Field	Value
Name	PNS ENTERPRISES
GSTIN	06ABJFP8733H1ZW
Address	Plot No. 2, Killa No. 25... Sonipat, Haryana - 131028
Contact	Akshay Sharma — 8395054056
Contact	Vinay Rajput — 8826816791

Config file: `frontend/src/config/company.ts`

Cloud accounts (who owns what)

Service	Typical login	Notes
GitHub	vrajput032	Repo owner
Cloudflare	(your CF login)	Pages project tallypns
Render	PNS workspace	Service tallypns-api
Supabase	(your Supabase login)	Project ref bwbbowajgazuljbxdownq

5. Environment files (what each means)

Rule

File	Commit to Git?
*.env.example	Yes (templates)
backend/.env	Never
frontend/.env	Never
frontend/.env.production	Never (gitignored)

backend/.env (local + same vars on Render)

Variable	Purpose	Your setup (safe summary)
DATABASE_URL	Prisma pooler connection	Supabase pooler port 6543 , pgbouncer=true
DIRECT_URL	Migrations & pg_dump	Same host port 5432 (direct/session)
JWT_SECRET	Sign access tokens (15m)	Long random string (local only)
JWT_REFRESH_SECRET	Sign refresh tokens (7d)	Different long random string
SUPABASE_URL	Project URL	https://bwbbowajgazuljbxdownq.supabase.co
SUPABASE_KEY	Supabase anon/API key	From Supabase → Settings → API
PORT	Express listen port	4000 (Render sets its own PORT)
RENDER_DEPLOY_HOOK_URL	Trigger backend deploy from CLI	From Render → Settings → Deploy Hook

Important:

- Use **DIRECT_URL (5432)** for dumps/migrations.
- Use **DATABASE_URL (6543)** for the running app.
- Never use pooler URL with `pg_dump`.

frontend/.env (local only)

```
VITE_API_URL=http://localhost:4000/api
```

frontend/.env.production (production builds)

```
VITE_API_URL=https://tallypns-api.onrender.com/api
```

Also locked in code: `frontend/src/lib/apiBaseUrl.ts` — production never uses localhost.

6. Frontend tech (learn these)

Piece	Library	Why
UI framework	React 19	Components & SPA
Build tool	Vite 8	Fast dev server & build
Language	TypeScript	Typesafety
Routing	react-router-dom 7	Pages / nested layout
Server state	TanStack Query	API fetch/cache
Tables	TanStack Table	Data grids
Forms	react-hook-form + Zod	Validation
HTTP	Axios	API client + JWT interceptors
Client state	Zustand	Auth tokens in memory/storage
Styling	Tailwind CSS 4	Utility CSS
Components	shadcn/ui + Base UI	Buttons, dialogs, sheets
Icons	lucide-react	Icons
Toasts	sonner	Notifications
Charts	recharts	(available for reports)

Frontend folder map

frontend/src/		
App.tsx		# Routes
features/		# One folder per business module
components/layout/		# AppShell, Sidebar, Topbar, PageHeader
components/ui/		# shadcn primitives
lib/api.ts		# Axios instance
lib/apiBaseUrl.ts		# Prod vs local API URL lock
store/authStore.ts		# Login state
config/company.ts		# Invoice header company info

Feature pages

Dashboard, Sales, Purchase, Inventory, Cash*, Bank*, GST, Customers, Vendors, Products, Reports
*Cash/Bank = "coming soon" UI stubs.

7. Backend tech (learn these)

Piece	Library	Why
HTTP server	Express 4	REST API
ORM	Prisma 6	Models + SQL

Validation	Zod	Request body schemas
Auth	jsonwebtoken	JWT
Passwords	bcryptjs	Hash passwords
Config	dotenv	Load .env
CORS	cors	Allow Pages → API

Backend folder map

```
backend/src/
  index.ts           # Listen 0.0.0.0:PORT
  app.ts             # Express + CORS + /health + /api
  routes/index.ts    # Mount all routers
  modules/*/         # auth, customers, sales, ...
  middleware/auth.ts # requireAuth (Bearer JWT)
  lib/prisma.ts      # Prisma client
backend/prisma/
  schema.prisma      # Database models
  migrations/        # SQL migrations
  seed.ts            # Admin user
```

Main API routes (all under /api)

Method	Path	Auth	Purpose
POST	/auth/login	No	Login
POST	/auth/refresh	No	New access token
GET	/dashboard/summary	Yes	Counts & stock value
CRUD	/customers	Yes	Customers
CRUD	/products	Yes	Products
CRUD	/vendors	Yes	Vendors
GET/POST/DELETE	/sales	Yes	Invoices
GET/POST/DELETE	/purchase	Yes	Bills
GET	/inventory/stock	Yes	Stock list
GET	/inventory/movements	Yes	Stock history
POST	/inventory/adjustments	Yes	Adjust stock
GET	/gst/summary	Yes	GST totals
GET	/reports/profit-loss	Yes	P&L
GET	/reports/stock	Yes	Stock report
GET	/cash, /bank	Yes	Stub 501
GET	/health	No	Render health check

Auth header: Authorization: Bearer <accessToken>

8. Database (Prisma models)

PostgreSQL on Supabase. Schema: `backend/prisma/schema.prisma`

Model	Purpose
User	Login accounts (ADMIN / STAFF)
Customer	Buyers
Vendor	Suppliers
Product	Items + stock + GST rate
SalesInvoice + SalesInvoiceItem	Sales with lines
PurchaseBill + PurchaseBillItem	Purchases with lines
StockMovement	IN / OUT / ADJUSTMENT
LedgerEntry	Future cash/bank ledger

Useful commands

```
npm run prisma:migrate -w backend    # apply migrations
npm run prisma:studio -w backend     # browse tables in browser
npm run seed -w backend              # create admin user
```

9. Daily commands (cheat sheet)

```
# Install once
npm install

# Local development (both servers)
npm run dev
# or separately:
npm run dev:backend
npm run dev:frontend

# Production deploy (ONE go – preferred)
npm run deploy

# Database safety
npm run db:backup           # full .dump restoreable backup
npm run db:export:sheets   # CSV for Excel / Google Sheets
npm run db:restore -- backups/tallypns-latest.dump
```

10. Deploy policy (save Render limits)

Render free tier is **monthly**:

- ~500 **pipeline minutes** (builds)
- ~750 **instance hours** (runtime)
- Free service **sleeps** when idle → first request may take 30–60s

Rule: finish frontend + backend work, then deploy **once**:

```
npm run deploy
```

Docs: `docs/DEPLOYMENT.md` , `docs/RELEASE_CHECKLIST.md`

Cursor rule: `.cursor/rules/deployment.mdc`

11. Backups (best practice)

Method	Command	Restorable?
Full dump (best)	<code>npm run db:backup</code>	Yes → new Postgres/Supabase
CSV / Excel	<code>npm run db:export:sheets</code>	Readable only, not full restore

Files go to `backups/` (**gitignored**). Copy dumps to Drive / iCloud / USB.

12. Common problems & fixes

Symptom	Cause	Fix
Mobile: cannot reach server	Frontend built with localhost API	<code>npm run deploy:frontend</code>
GST / Inventory 404	Render on old commit	Redeploy backend / <code>npm run deploy</code>
Invalid email/password (but password OK)	Network/API down OR wrong creds	Check <code>/health</code> , then seed admin
Slow first load	Render cold start	Wait & retry
<code>pg_dump</code> fails	Using pooler port 6543	Use <code>DIRECT_URL</code> port 5432

13. Mental model for learning

1. **Frontend** = UI + calls `/api`
2. **Backend** = business rules + JWT + Prisma
3. **Supabase** = only the database (not hosting the Node app)
4. **Cloudflare** = static React files
5. **Render** = Node API process
6. **Env files** = secrets & URLs for each environment
7. **Deploy once** = protect free-tier minutes

When you change a feature: update **backend module** + **frontend feature** + test locally → one `npm run deploy` .

14. Where to look in code

I want to...	Open
Add a page	<code>frontend/src/App.tsx</code> + <code>features/...</code>
Add an API	<code>backend/src/modules/...</code> + <code>routes/index.ts</code>
Change DB table	<code>backend/prisma/schema.prisma</code> then migrate
Change login	<code>auth modules</code> + <code>LoginPage.tsx</code>
Change invoice header	<code>frontend/src/config/company.ts</code>
Change API URL logic	<code>frontend/src/lib/apiBaseUrl.ts</code>

Deploy script	scripts/deploy.sh
---------------	-------------------

End of guide — PNS ERP / TallyPNS learning reference